

TANUSH THAKRAN

AI / ML Engineer

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SUMMARY

Machine learning engineer who ships models with evidence behind them: leakage-safe evaluation, controlled ablations, and metrics reported against honest baselines. Experience spans learning-to-rank on multi-million-row datasets, applied ML with scikit-learn, and LLM/RAG systems, owning the full path from data extraction through deployment.

TECHNICAL SKILLS

Machine Learning: LightGBM, scikit-learn, Feature Engineering, Cross-Validation, Model Evaluation. **Familiar with:** PyTorch, TensorFlow
LLM & Retrieval: RAG, Embeddings, Vector Search (ChromaDB), Prompt Engineering, LLM Evaluation, Groq API, Gemini API, Whisper, Ollama
Data & Analytics: Python, SQL, PostgreSQL, Pandas, NumPy, PyArrow/Parquet, Power BI, DAX
Web & Infra: FastAPI, Node.js, React.js, Celery, Redis, Docker, Streamlit, pytest, GitHub Actions, Git, Flutter

MACHINE LEARNING & DATA PROJECTS

Pokémon TCG Move Prediction — Learning-to-Rank

Python, LightGBM, PyArrow | github.com/Tanush1206/pokemon-tcg-move-prediction

- Framed move selection as learning-to-rank (not classification): LGBMRanker with a LambdaRank objective on 3.35M rows and 83 features, reaching 57.4% top-1 accuracy on 50,649 held-out decisions vs. a 21.7% random baseline (2.65x); true move in top 3 88% of the time.
- Ran a 7-run ablation study with episode-level, leakage-safe splits; the largest single gain (+0.054 accuracy) came from reverse-engineering undocumented option indices to recover card identity, while filtering for “stronger” episodes did not help.

Customer Churn Prediction — Deployed ML App

Python, scikit-learn, Streamlit | github.com/Tanush1206/Customer-Churn-Prediction

- Built an end-to-end telecom churn pipeline (leakage-safe ColumnTransformer; Logistic Regression, Random Forest, Gradient Boosting compared) deployed as a Streamlit app; kept Logistic Regression (within 0.003 ROC-AUC of Gradient Boosting, interpretable) and set a 0.30 threshold by maximising F1 on out-of-fold predictions, lifting test recall from 55.9% to 75.1% at 51.8% precision (84.1% ROC-AUC).

RAG-Based AI — Offline Retrieval-Augmented QA Pipeline

Python, Whisper, Ollama | github.com/Tanush1206/rag_based_ai

- Built a fully offline RAG pipeline over Hindi-language course videos (Whisper large-v2 transcription and translation, local bge-m3 embeddings, top-5 cosine retrieval, deepseek-r1 via Ollama) with answers cited to video, title, and timestamp; zero external API calls.
- Tuned chunk size and overlap on a manually labelled question set; fixed a scale-only embedding crash (oversized requests) with fixed-size batching.

LLM Utility Lab

Python, Groq API, FastAPI, Docker | github.com/Tanush1206/llm-utility-lab

- Built a modular LLM application (client, prompting, summarization, context-aware Q&A, evaluation) served as a FastAPI service in Docker on Render, with per-request token tracking, an evaluation runner that flags pass-to-fail regressions, and 58 pytest tests in GitHub Actions CI.

Superstore Profitability Analysis — SQL & Power BI

SQL, PostgreSQL, Power BI, DAX | github.com/Tanush1206/superstore-powerbi-analysis

- On the public Sample Superstore dataset (9,994 line items, \$2.3M revenue), bucketed sales by discount band in SQL — margin turns negative past 25% discount, reaching ~77% beyond 40% — and shipped a Power BI dashboard (DAX, cohort/RFM segmentation) with a recommendation estimated at ~\$35K recoverable annual profit, assuming order volume holds.

VideoCaptionMaker — AI Video Captioning App (Feature-Complete, Pre-Deployment)

Python, FastAPI, Celery, ChromaDB | github.com/Tanush1206/Video-Caption-Maker

- Built a GPU-accelerated faster-whisper transcription service across a six-service Docker Compose architecture (Next.js, FastAPI, Celery, Redis, PostgreSQL, ChromaDB) that keeps long inference off the request path, with ChromaDB semantic search, Gemini-backed Q&A, and 188 backend tests.

PactPal — LLM Contract Simplifier

React, Node.js/Express, Gemini 2.5 Flash (Vertex AI) | github.com/Tanush1206/PactPal-Frontend

- Built a web app that turns contracts into a plain-English guide section by section with Gemini 2.5 Flash, with clause-level explanations.

EXPERIENCE

App Developer Intern — CYPARTA

Dec 2025 – Jan 2026

Cairo, Egypt (Remote) | Flutter, Firebase

- Built the Driver & Medic mobile app end-to-end, handling real-time GPS telemetry and state consistency across concurrent users for live logistics.
- Implemented trip-lifecycle workflows and post-trip EMS reporting modules for data integrity.

Junior Manager — AIESEC

Oct 2025 – Mar 2026

Bengaluru, India | Managed Global Volunteer operations, tracked departmental KPIs, led cross-functional teams.

EDUCATION

Integrated B.Sc. in Computer Science — BITS Pilani

2024 – 2028

CGPA: 7.64 | Coursework: Algorithms, Data Structures, Database Systems, Operating Systems, Computer Networks, System Design, Statistics, Deep Learning

CERTIFICATIONS

Machine Learning: Advanced Learning Algorithms — Stanford / DeepLearning.AI • Supervised ML: Regression & Classification — Stanford / DeepLearning.AI • Intermediate Machine Learning — Kaggle

LLM & GenAI: LangChain: Chat with Your Data (LangChain, DeepLearning.AI) • Building Systems with the ChatGPT API (OpenAI, DeepLearning.AI) • ChatGPT Prompt Engineering for Developers (OpenAI, DeepLearning.AI) • AI Fluency: Framework & Foundations (Anthropic)